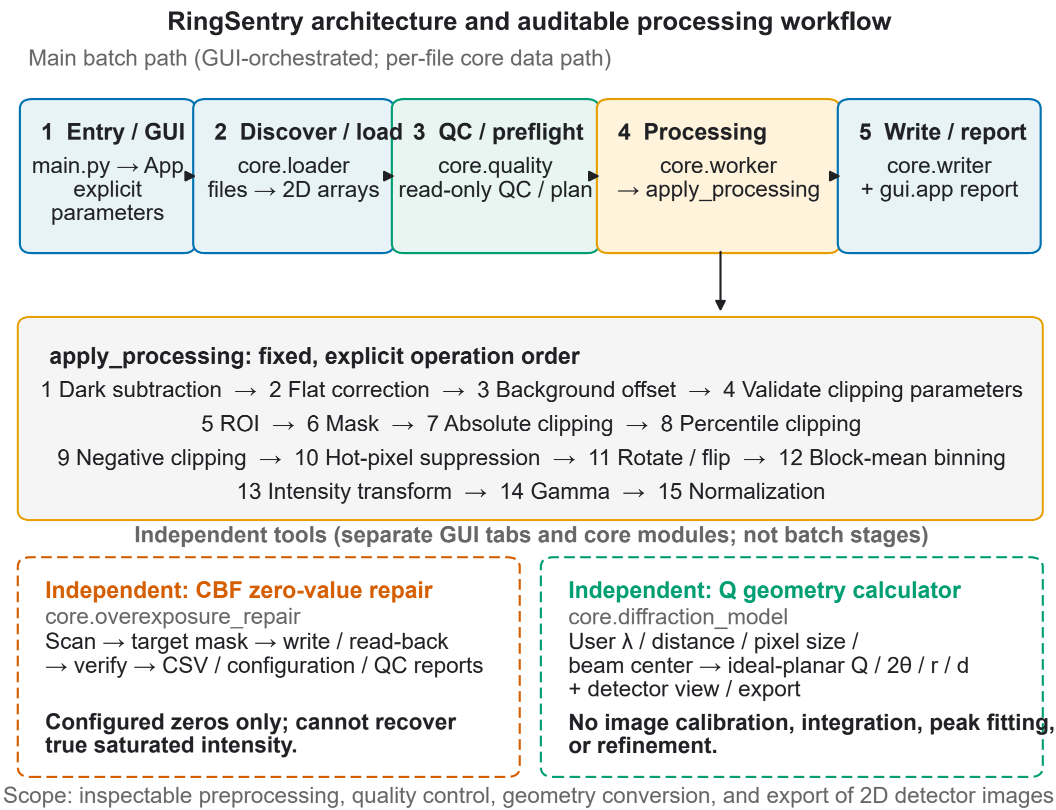
RingSentry architecture and auditable processing workflow
Main batch path (GUI-orchestrated; per-file core data path)
1 Entry / GUI
main.py → App
explicit parameters
2 Discover / load
core.loader
files → 2D arrays
3 QC / preflight
core.quality
read-only QC / plan
4 Processing
core.worker
→ apply_processing
5 Write / report
core.writer
+ gui.app report
apply_processing: fixed, explicit operation order
1 Dark subtraction → 2 Flat correction → 3 Background offset → 4 Validate clipping parameters
5 ROI → 6 Mask → 7 Absolute clipping → 8 Percentile clipping
9 Negative clipping → 10 Hot-pixel suppression → 11 Rotate / flip → 12 Block-mean binning
13 Intensity transform → 14 Gamma → 15 Normalization
Independent tools (separate GUI tabs and core modules; not batch stages)
Independent: CBF zero-value repair
core.overexposure_repair
Scan → target mask → write / read-back
→ verify → CSV / configuration / QC reports
Configured zeros only; cannot recover true saturated intensity.
Independent: Q geometry calculator
core.diffraction_model
User λ / distance / pixel size /
beam center → ideal-planar Q / 2θ / r / d
+ detector view / export
No image calibration, integration, peak fitting, or refinement.
Scope: inspectable preprocessing, quality control, geometry conversion, and export of 2D detector images.